

APPLICATION SECTION

SCHEDULING OF PATROLS TO GUARD AN INSTALLATION

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Summary: In this paper we propose a method of guarding an
 installation with random patrol times. Expressions
 are derived for the probability of catching an intruder
 for this method as well as for the method with fixed
 patrol times. It is shown how these results can be
 used as a decision making aid in the allocation of
 manpower.

1. INTRODUCTION

Consider an installation that has to be guarded against possible 'intruders'. Insufficient manpower is available to guard it continuously. The question for the decision maker is how many patrols are needed and how they should be scheduled to attain a given probability of catching an intruder. To be more specific, suppose the installation has to be guarded for a period of T minutes, typically taken as night-time. Suppose also that enough manpower is available to supply patrols for a total of N minutes, with N usually much smaller than T . There are clearly many different ways of allocating the available time. Perhaps the most frequently used method is that of constant time between patrols. The concern of our client about this method is that an intruder could possibly observe the regularity of the patrols and thus try to gain access to the installation during an unpatrolled

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period. An alternative would be to schedule the patrols in some random fashion.

In this paper we propose a particular random patrol method that suited our particular application. For this method as well as for the method of fixed patrol times we derive the probabilities of catching an intruder for two different situations. These derivations rely on results for order statistics from a finite population and may be of independent interest.

In Section 2 the methods are defined and in Section 3 the probabilities of catching an intruder are derived. In Section 4 it is shown how these results were used as a decision making aid and some numerical results are given.

2. PATROL METHODS

As mentioned in Section 1, let T be the time (in minutes) that the installation has to be guarded and suppose that N minutes are available for patrol time. Two cases are distinguished. Case I is where the intruder does not observe the guard and we assume that he commences his act at a time which is uniformly distributed over the interval $(0, T)$. Case II is where he can observe the guard and will thus not commence his act while there is a guard on patrol. This is the case that we are particularly interested in.

We will further assume that there are n patrols each of duration s minutes (ie $N = ns$). We now describe the fixed patrol and random patrol methods.

Method 1 (Fixed patrol times).

Divide the interval $(0, T)$ into n subintervals of equal length, say $\ell = T/n$ (assumed to be an integer). The first s minutes in each subinterval are patrolled. The patrol times are then the intervals

$$((i-1)\ell, (i-1)\ell + s] \quad i = 1, 2, \dots, n$$

Method 2 (Random patrol times).

For this method the total time T is divided into intervals each of length s and from these intervals the required number of patrol periods are randomly chosen. To be more precise, divide the interval $(0, T)$ into $A = T/s$ (assumed to be an integer) subintervals, each of length s . Choose n intervals at random from A and patrol during the chosen intervals. Let

$g_1, g_2, \dots, g_n$ be n numbers chosen at random, without replacement from $\{1, 2, \dots, A\}$. The patrol times are then the intervals

$$((g_i - 1)s, g_i s] \quad i = 1, 2, \dots, n.$$

Note that in the application of this method we first order the chosen numbers from small to large and denote by $1 \leq g_{(1)} < g_{(2)} < \dots < g_{(n)} \leq A$ these ordered numbers. The consecutive patrol intervals are then

$$((g_{(i)} - 1)s, g_{(i)} s] \quad i = 1, 2, \dots, n.$$

3. CATCH PROBABILITIES

In this section we derive expressions for the probabilities of catching an intruder. We denote by t the time (in minutes) that it takes an intruder to complete his act. We first consider case I. Let $p_i(T, n, s, t)$, $i = 1, 2$ be the probability of catching an intruder if patrol method i is used.

In the Appendix (Theorem 1) we show that in this case the following expressions hold for $p_i(T, n, s, t)$, $i = 1, 2$.

$$\begin{aligned} p_1(T, n, s, t) &= n(s+t)/T && \text{if } n \leq T/(t+s), \\ &= 1 && \text{otherwise} \end{aligned}$$

and

$$\begin{aligned} p_2(T, n, s, t) \\ = 1 - \binom{A}{n}^{-1} \left\{ (1-v) \binom{A - ([t/s] + 1)}{n} + v \binom{A - ([t/s] + 2)}{n} \right\} \end{aligned}$$

where $v = t/s - [t/s]$, with $[.]$ the integer function.

Remarks

1. Note that the formula for $p_2(T, n, s, t)$ is intuitively "almost" obvious, at least for v close to zero. For $v = 0$ the term in curly brackets denotes the probability of choosing n objects from A such that $[t/s] + 1$ specific objects are not included.
2. If $t < s$ then $[t/s] = 0$ and the probability becomes

$$p_2(T, n, s, t) = 1 - \left\{ (1 - t/s) \frac{A - n}{A} + \frac{t}{s} \frac{(A - n)(A - n - 1)}{A(A - 1)} \right\}.$$

We now consider case II. In this case the intruder observes the guard and we assume that he starts his act as soon as a patrol interval ends.

In the Appendix (Theorem 2) we show that in this case the following results hold for $p_i(T, n, s, t)$, $i = 1, 2$.

$$\begin{aligned} p_1(T, n, s, t) &= 1 \quad \text{if } n \geq T(t + s)^{-1} \\ &= 0 \quad \text{if } n < T(t + s)^{-1} \end{aligned}$$

and

$$p_2(T, n, s, t) = 1 - \binom{A}{n}^{-1} \binom{A - ([t/s] + 1)}{n}.$$

Remarks

1. As mentioned before, there are many possible ways of carrying out random patrols. The one considered in this paper was the one preferred by our client. Another possibility would be the following. Let $U_1, U_2, \dots, U_n$ be independent, identically distributed random variables from the uniform distribution on $(0, T - ns)$. Denote by $U_{(1)} < U_{(2)} < \dots < U_{(n)}$ the corresponding order statistics. Take as patrol times the intervals

$$(U_{(i)} + (i - 1)s, U_{(i)} + is], \quad i = 1, 2, \dots, n.$$

Note that the gaps between two consecutive patrol intervals are the same as the gaps between the corresponding order statistics, i.e.

$$(U_{(i+1)} + is) - (U_{(i)} + is) = U_{(i+1)} - U_{(i)}.$$

By using the known results for the distribution of uniform spacings, expressions for the relevant probabilities for this method can be written down for both cases I and II.

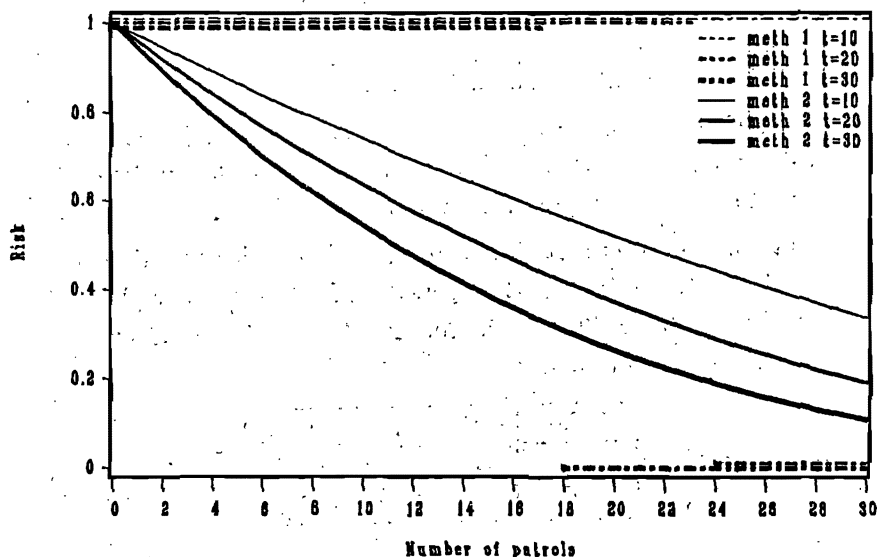
2. Some results on the distribution of order statistics from a finite population were recently obtained by Meyer (1987). His results and area of application are, however, different from ours.

4. NUMERICAL RESULTS

The results of Section 3 were used by our client to compare the relative advantages of fixed and random patrol methods. Furthermore, he used it as a decision making aid in the following way. For a given risk (probability of not catching an intruder), he used the results to see how many patrols were needed in order not to exceed that risk. Alternatively, in cases where his number of patrols is restricted due to manpower shortages and/or budgetary constraints, the results were used to quantify his risk and transfer manpower and/or funds from other sources to bring down this risk if it became too high.

For user friendly practical application of the results, the client was supplied with graphs of number of patrols versus risk for those cases that were of concern to him. These graphs were of the format shown in Figure 4.1 below.

Figure 4.1 Risk as a function of number of patrols for case II with $T = 720, s = 10, t = 10, 20, 30$.



The range of parameter values of interest was $T: 360(30)840$ (i.e. 6 to 14 hours), $s: 1, 5, 10$, $t: 10, 20, 30$.

Although the full results are not reproduced, we make a number of comments based on them.

(i) For case I, method 1 is always better than method 2, although typically not much better. This is in line with one's intuition and says that if the arrival time of the intruder is assumed to be uniformly distributed then nothing is gained by random patrols.

(ii) In case II the situation is of course completely different with method 1 now having a 0-1 risk and method 2 being better whenever method 1 is not 0. From this and the fact that in our application the client was uncertain whether the intruder could see the guard or not, one should opt for method 2.

(iii) If t is overestimated the risks can be significantly larger than anticipated. Consider eg case II with $t = 720$, $s = 1$, $n = 15$. Then we have the following risks

Method	$t=30$	20	10
1	1	1	1
2	0.51	0.64	0.79

Thus if one estimates $t = 30$ instead of 20 then one could think the risk is 0.51 while it really is 0.64. One should thus be conservative in the estimation of t .

(iv) For fixed values of T, t, n , the risks do not decrease dramatically as s is increased (for method 2). As an example, take $T = 720$, $t = 30$, $n = 15$. The risks for the choices of s of interest are then as follows

Method	$s=1$	5	10
1	1	1	1
2	0.51	0.45	0.38

(v) If the total patrol time $N = ns$ is fixed, it is better to have a large number of short patrol intervals rather than a few long ones. For the case $T = 720, t = 30, ns = 20$ we have for example the following risks for method 2

$n=20, s=1$	$n=4, s=5$
0.41	0.82

In practice, however, there are usually other factors that make it difficult to have too large a number of patrols.

(vi) The results just discussed were used by our client as a decision making aid in his particular situation. Other situations might have other factors playing a role, which would then need different models. Further work could be done along these lines.

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APPENDIX

Here we prove the results used in Section 3.

Theorem 1. For case I we have

$$p_1(T, n, s, t) = \begin{cases} n(s+t)/T & \text{if } n \leq T/(t+s) \\ 1 & \text{otherwise} \end{cases}$$

and

$$p_2(T, n, s, t) = 1 - \binom{A}{n}^{-1} \left\{ (1-v) \binom{A - ([t/s] + 1)}{n} + v \binom{A - ([t/s] + 2)}{n} \right\}$$

where $v = t/s - [t/s]$, with $[\cdot]$ the integer function.

Proof. Let X denote the time at which the intruder starts his act. Then by assumption X has a $U(0, T)$ distribution. By writing

$$\begin{aligned} p_1(T, n, s, t) &= \sum_{i=1}^n P(\text{intruder is caught in patrol interval } i) \\ &= \sum_{i=1}^n \int_0^T T^{-1} I(\max(il - s - t, (i-1)\ell) < x \leq il) dx, \end{aligned}$$

with $I(\cdot)$ the indicator function, the result for p_1 follows in a straightforward manner.

Now consider method 2. Let again X denote the start of the intruder's act. The patrol times are the intervals

$$((g_i - 1)s, g_i s]$$

with $g_1, g_2, \dots, g_n$ a sample taken without replacement from $\{1, 2, \dots, A\}$.

Given $X = x$, the intruder is caught if and only if

$$(g_i - 1)s - t < x \leq g_i s,$$

for at least one $1 \leq i \leq n$.

This is equivalent to

$$x/s \leq g_i < (x+t)/s + 1,$$

for at least one $1 \leq i \leq n$.

Note also that g_1 is uniformly distributed over $\{1, 2, \dots, A\}$ and g_i given $g_1, g_2, \dots, g_{i-1}$ is uniformly distributed over $\{1, 2, \dots, A\} \setminus \{g_1, g_2, \dots, g_{i-1}\}$. Using these facts and defining $I = I(x, s, t) \equiv [x/s, (x+t)/(s+1)]$, we can write

$$\begin{aligned} p_2(T, n, s, t) &= \int_0^T T^{-1} P(g_i \in I; \text{for at least one } 1 \leq i \leq n) dx \\ &= 1 - \int_0^T T^{-1} P(g_i \notin I; \text{for all } 1 \leq i \leq n) dx \\ &= 1 - T^{-1} \int_0^T P(g_1 \notin I) \prod_{j=2}^n P(g_j \notin I | g_1 \notin I, g_2 \notin I, \dots, g_{j-1} \notin I) dx \\ &= 1 - T^{-1} \int_0^T (1 - A^{-1} C(x, s, t)) \prod_{j=2}^n (1 - (A-j+1)^{-1} C(x, s, t)) dx \\ &= 1 - T^{-1} \int_0^T \prod_{j=1}^n (1 - (A-j+1)^{-1} C(x, s, t)) dx \end{aligned}$$

where $C(x, s, t)$ denotes the number of integers included in the interval I_x . In order to evaluate this integral we have to split the range of integration into subintervals over which $C(x, s, t)$ can be calculated explicitly. To this end, let $v = t/s - [t/s]$ be the fractional part of t/s . Note that in typical applications $t > s$. Split the interval $(0, T)$ into subintervals

$$((i-1)s, is - vs], (is - vs, is] \quad i = 1, 2, \dots, A.$$

Consider first the interval $(i-1)s < x \leq is - vs$. On this interval $i-1 < x/s \leq i-v$, and thus the smallest integer greater than or equal to x/s is i . Also, on this interval the largest integer strictly less than $(x+t)/s + 1$ is $i + [t/s]$, since $i + [t/s] + v < (x+t)/s + 1 \leq i + [t/s] + 1$. Thus on this interval $C(x, s, t) = [t/s] + 1$.

Consider now the interval $is - vs < x \leq is$. Here $i-v < x/s \leq i$ and the smallest integer greater than or equal to x/s is i . Also, $i + [t/s] + 1 < (x+t)/s + 1 \leq i + [t/s] + 1 + v$, and the largest integer strictly less than $(x+t)/s + 1$ is $i + [t/s] + 1$. Thus on this interval $C(x, s, t) = [t/s] + 2$.

Using these, we can write the above integral as

$$\begin{aligned} & \int_0^T \prod_{j=1}^n (1 - (A-j+1)^{-1} C(x, s, t)) dx \\ &= \sum_{i=1}^A \left[\int_{(i-1)s}^{is-vs} \prod_{j=1}^n \left(1 - \frac{[t/s]+1}{A-j+1} \right) dx + \int_{is-vs}^{is} \prod_{j=1}^n \left(1 - \frac{[t/s]+2}{A-j+1} \right) dx \right] \\ &= T \left\{ (1-v) \prod_{j=1}^n \left(1 - \frac{[t/s]+1}{A-j+1} \right) + v \prod_{j=1}^n \left(1 - \frac{[t/s]+2}{A-j+1} \right) \right\} \\ &= T \left\{ (1-v) \binom{A}{n}^{-1} \binom{A - [t/s] - 1}{n} \right. \\ & \quad \left. + v \binom{A}{n}^{-1} \binom{A - [t/s] - 2}{n} \right\}. \end{aligned}$$

The result follows.

Theorem 2. For case II we have

$$\begin{aligned} p_1(T, n, s, t) &= 1 \quad \text{if } n \geq T(t+s)^{-1} \\ &= 0 \quad \text{if } n < T(t+s)^{-1} \end{aligned}$$

and

$$p_2(T, n, s, t) = 1 - \binom{A}{n}^{-1} \binom{A - [t/s + 1]}{n}.$$

Proof. The expression for p_1 is immediate. Now consider p_2 .

Let $1 \leq g_{(1)} < g_{(2)} < \dots < g_{(n)} \leq A$ denote the ordered g 's. It is clear that

$$p_2(T, n, s, t) = P((g_{(i+1)} - g_{(i)} - 1)s \leq t),$$

if he chooses to start his act immediately after the end of the i -th patrol interval (we see below that the distributions of $g_{(i+1)} - g_{(i)}$ are the same for all i).

Thus in order to find an expression for p_2 we need to find the distribution of the spacings $\{g_{(i+1)} - g_{(i)}\}$. We now derive this.

Let $\underline{g} = (g_1, g_2, \dots, g_n)$ and $\underline{R} = (g_{(1)}, g_{(2)}, \dots, g_{(n)})$,

then

$$\begin{aligned} P(\underline{g} = \underline{\gamma}) &= \{n! \binom{A}{n}\}^{-1} && \text{if } \gamma_i \in \{1, 2, \dots, A\} \\ &&& \text{and } \gamma_i \neq \gamma_j, i \neq j \\ &= 0 && \text{otherwise} \end{aligned}$$

from which follows

$$\begin{aligned} P(\underline{R} = \underline{r}) &= \binom{A}{n}^{-1} && \text{if } 1 \leq r_1 < r_2 < \dots < r_n \leq A \\ &= 0 && \text{otherwise.} \end{aligned}$$

If we make the transformation

$$\begin{aligned} V_1 &= g_{(1)} \\ V_i &= g_{(i)} - g_{(i-1)}, \quad i = 2, 3, \dots, n, \end{aligned}$$

then it follows that

$$\begin{aligned} P(\underline{V} = \underline{v}) &= \binom{A}{n}^{-1} && \text{if } \sum_{i=1}^n v_i \leq A \text{ and } v_i \geq 1, \quad i = 1, 2, \dots, n \\ &= 0 && \text{otherwise.} \end{aligned}$$

Since this distribution is symmetric in $V_1, V_2, \dots, V_n$ it follows that the distribution of any particular $k < n$ of the spacings $V_1, V_2, \dots, V_n$ is the same as that of any other set of k of them. Thus, the distribution of

$V_{i+1} + \dots + V_j = g_{(j)} - g_{(i)}$ must be the same as that of $V_1 + \dots + V_{j-1} = g_{(j-1)}$. In particular, taking $j = i + 1$ gives

$$g_{(i+1)} - g_{(i)} \stackrel{D}{=} g_{(1)},$$

and thus all spacings have the same distribution. In order to find this distribution, we consider the marginal distribution of any $g_{(i)}$. To this end, we use the following

$$\begin{array}{ccccccc} \text{choose } i-1 \text{ from these} & & & & \text{choose } n-i \text{ from these} & & \\ \hline 1 & 2 & & k-1 & k & k+1 & A \end{array}$$

to see that

$$\begin{aligned} P(g_{(i)} = k) &= \binom{k-1}{i-1} \binom{A-k}{n-i} \binom{A}{n}^{-1} & \text{if } i \leq k \leq A - (n-i) \\ &= 0 & \text{otherwise.} \end{aligned}$$

In particular, this gives

$$P(g_{(i+1)} - g_{(i)} = k) = P(g_{(1)} = k) = \binom{A-k}{n-1} \binom{A}{n}^{-1}, \quad 1 \leq k \leq A - (n-1).$$

From this we obtain the distribution function of a spacing as

$$\begin{aligned} G(x) \equiv P(g_{(i+1)} - g_{(i)} \leq x) &= \sum_{k \leq [x]} P(g_{(i+1)} - g_{(i)} = k) \\ &= \binom{A}{n}^{-1} \sum_{k=1}^{[x]} \binom{A-k}{n-1}. \end{aligned}$$

Using (12.6), p63 in Feller (1968), we have

$$\begin{aligned} \sum_{k=1}^{[x]} \binom{A-k}{n-1} &= \binom{A+1}{n} - \binom{A-[x]}{n} - \binom{A}{n-1} \\ &= \binom{A}{n} - \binom{A-[x]}{n}, \end{aligned}$$

which gives

$$G(x) = 1 - \binom{A-[x]}{n} \binom{A}{n}^{-1}$$

The result for p_2 now follows from the fact that

$$p_2(T, n, s, t) = G(t/s + 1).$$

Remark

From the proof we see that the spacings in a sample without replacement from $\{1, 2, \dots, A\}$ all have the same distribution, which is the same as that of the first order statistic. This result is analogous to a similar result for a sample from a $U(0, 1)$ distribution (see eg David (1981)).